Data Analytics Internship

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Internship Company: Rooman Technologies

Domain: Data Analytics

Project: Data-Driven Insights into OLA Ride Bookings using Power BI

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Phase 2 – Build

1. Problem Statement

In ride-booking platforms like OLA, massive volumes of trip data are generated daily. However, extracting meaningful insights from this data is challenging. Traditional reports fail to highlight:

- Peak booking hours
- Revenue patterns
- Ride cancellations
- Demand-supply imbalance

This makes it difficult for decision-makers to optimize pricing, improve customer satisfaction, and maximize revenue.

2. Introduction

This project focuses on analyzing OLA ride booking data to extract actionable insights related to:

- Ride demand patterns
- Revenue generation
- Customer behavior
- Ride status distribution

The goal is to transform raw booking data into meaningful insights using **data analytics and visualization techniques**, enabling better operational and strategic decisions.

3. Tech Stack

- Python (Jupyter Notebook) – Data analysis
- Pandas, NumPy – Data cleaning & processing
- Matplotlib, Seaborn – Data visualization
- Power BI – Interactive dashboard

4. Data Cleaning

- Converted date/time columns into proper **datetime format**
- Extracted useful features like **hour, day, month**
- Removed duplicate records
- Handled missing/null values
- Standardized column names (lowercase, underscores)
- Verified dataset using .shape, .info(), .describe()

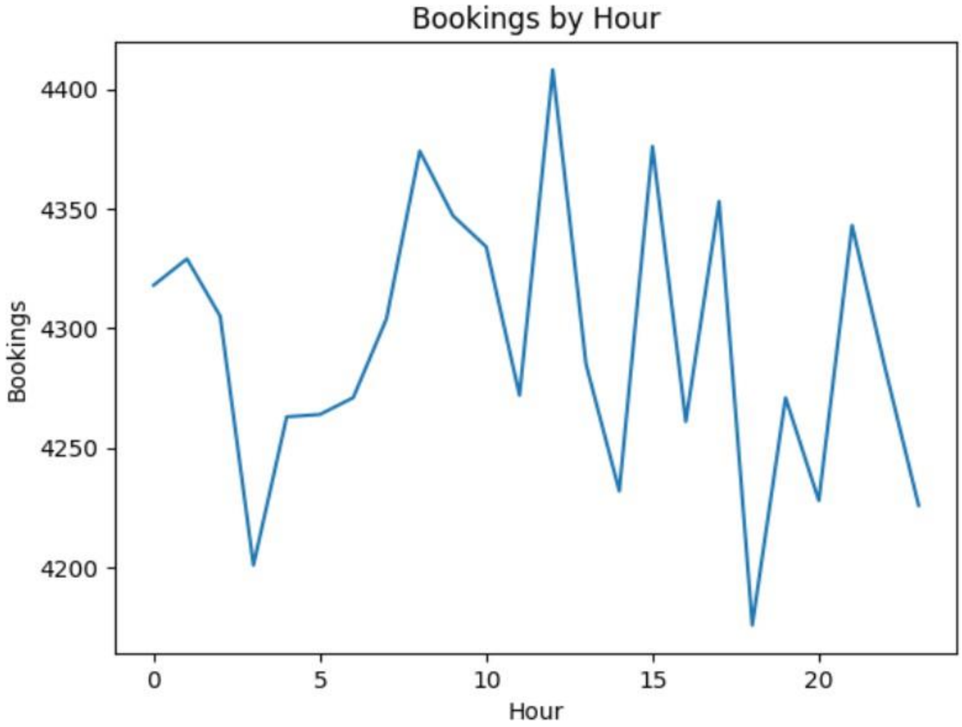
5. Exploratory Data Analysis (EDA)

- Analyzed booking distribution across different hours
- Studied revenue trends over days
- Examined ride status (Completed, Cancelled, etc.)
- Identified peak demand periods
- Evaluated customer and driver behavior patterns

6. Core Analysis & Visualizations

1. Bookings by Hour (Line/Bar Chart)

Shows ride demand variation throughout the day.

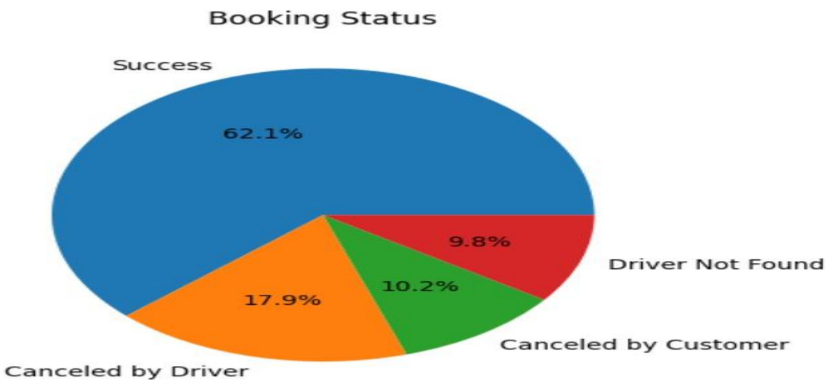


Insight:

Peak bookings occur during morning (office hours) and evening (rush hours), indicating high demand during commute times.

2. Ride Status Distribution (Pie Chart)

Shows proportion of Sucsess, Cancelled rides

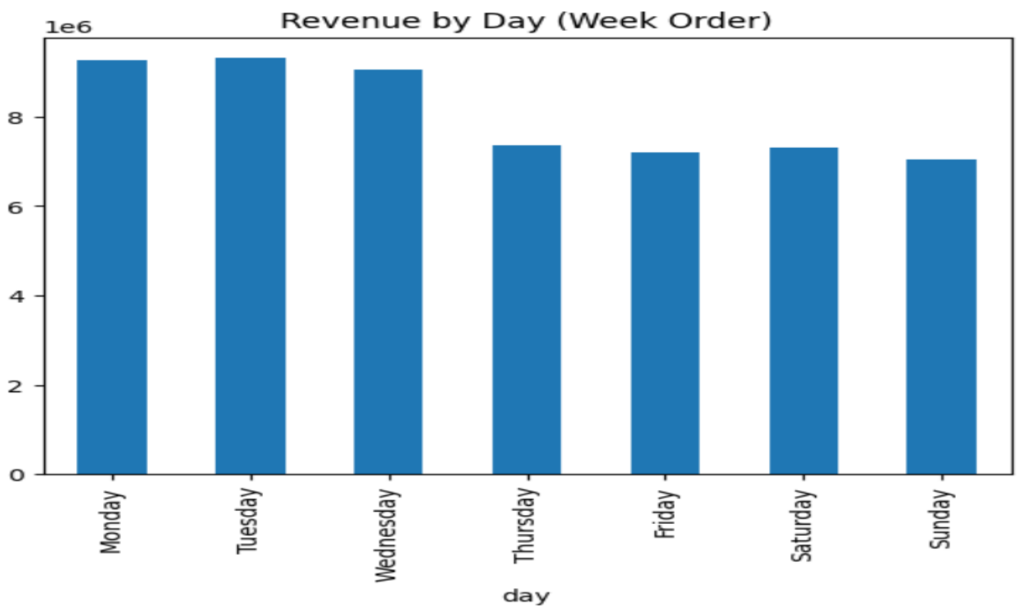


Insight:

A noticeable percentage of rides are cancelled, which highlights service inefficiencies or driver availability issues.

3. Revenue by Day (Bar Chart)

Displays total revenue generated each day.



7. Key Insights

- Peak ride demand occurs during **rush hours**
- Revenue is highly dependent on **booking volume**

- Certain locations dominate ride demand
- Ride cancellations impact overall efficiency
- Demand patterns show **time-based trends**

8. Statistical Validation

- Correlation analysis confirms:
    - Bookings ↑ → Revenue ↑
- Time-based grouping validates peak hour trends
- Distribution analysis confirms demand imbalance

9. Dashboard Development

Power BI Dashboard Features

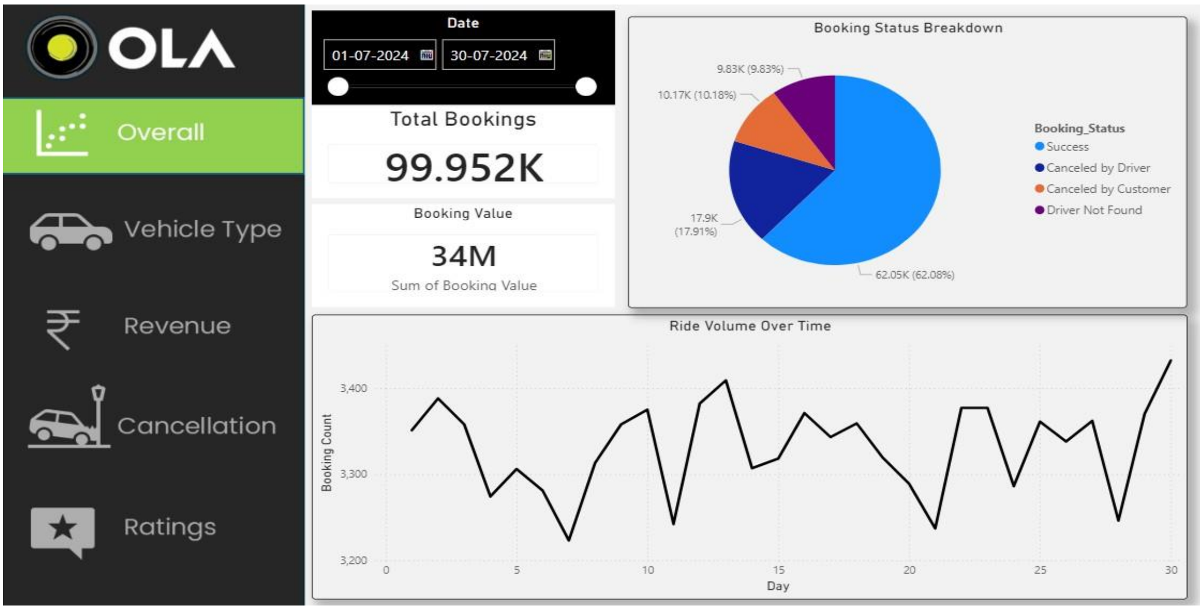
- KPI Cards:
    - Total Bookings
    - Total Revenue
    - Cancellation Rate
- Visualizations:
    - Bookings by Hour
    - Revenue Trends
    - Ride Status Distribution

- Location-based Demand

- Filters:
  - Date
  - Location
  - Ride Status

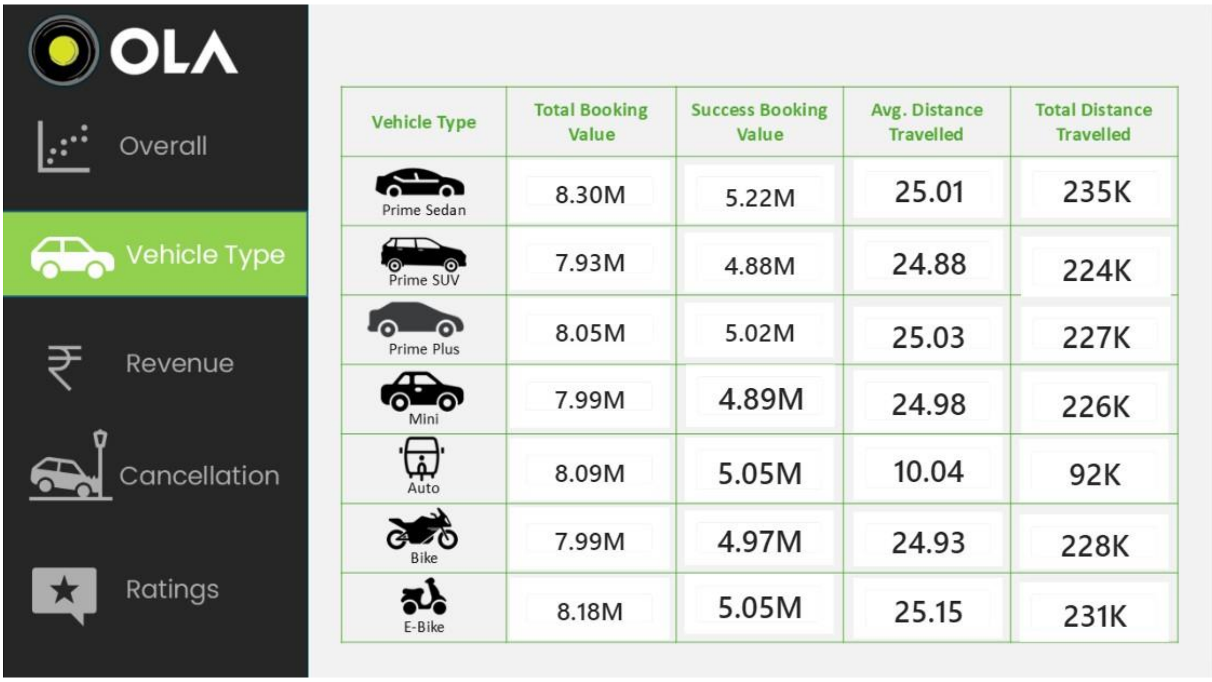
Ola Insights Dashboard (Power BI)

Dashboard Overview – OLA Booking Analysis



Displays bookings, revenue, ride trends, and cancellation status analysis.

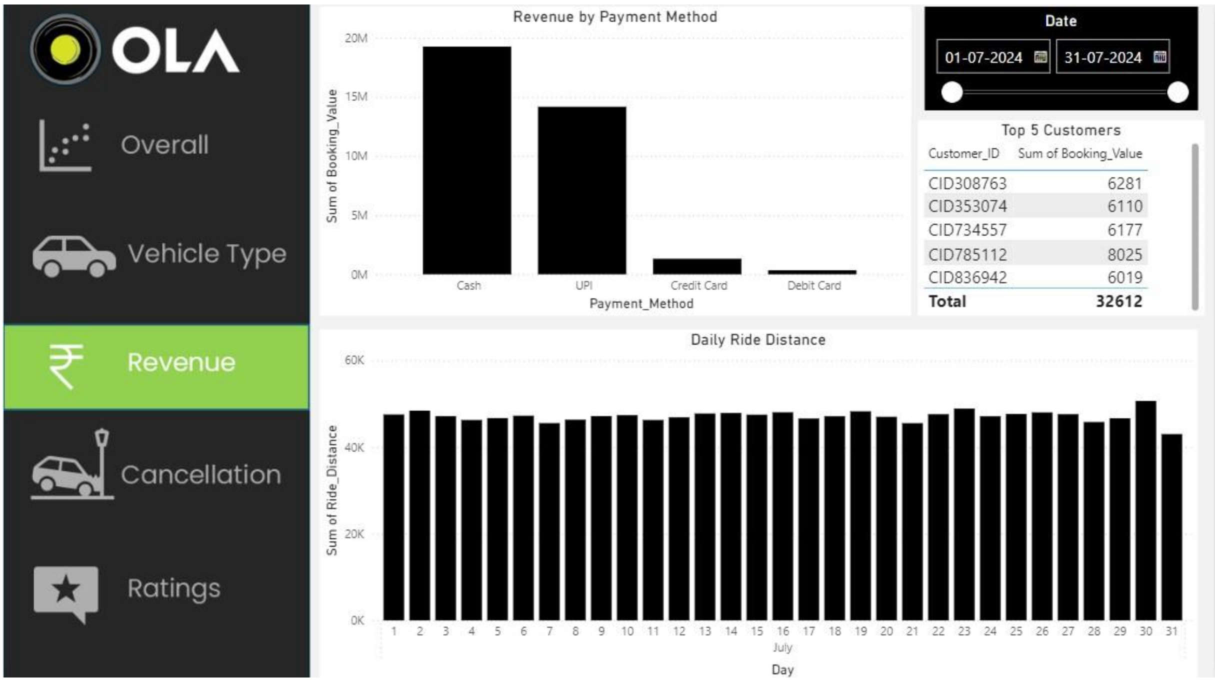
Vehicle Type Performance Analysis



Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance Travelled	Total Distance Travelled
Prime Sedan	8.30M	5.22M	25.01	235K
Prime SUV	7.93M	4.88M	24.88	224K
Prime Plus	8.05M	5.02M	25.03	227K
Mini	7.99M	4.89M	24.98	226K
Auto	8.09M	5.05M	10.04	92K
Bike	7.99M	4.97M	24.93	228K
E-Bike	8.18M	5.05M	25.15	231K

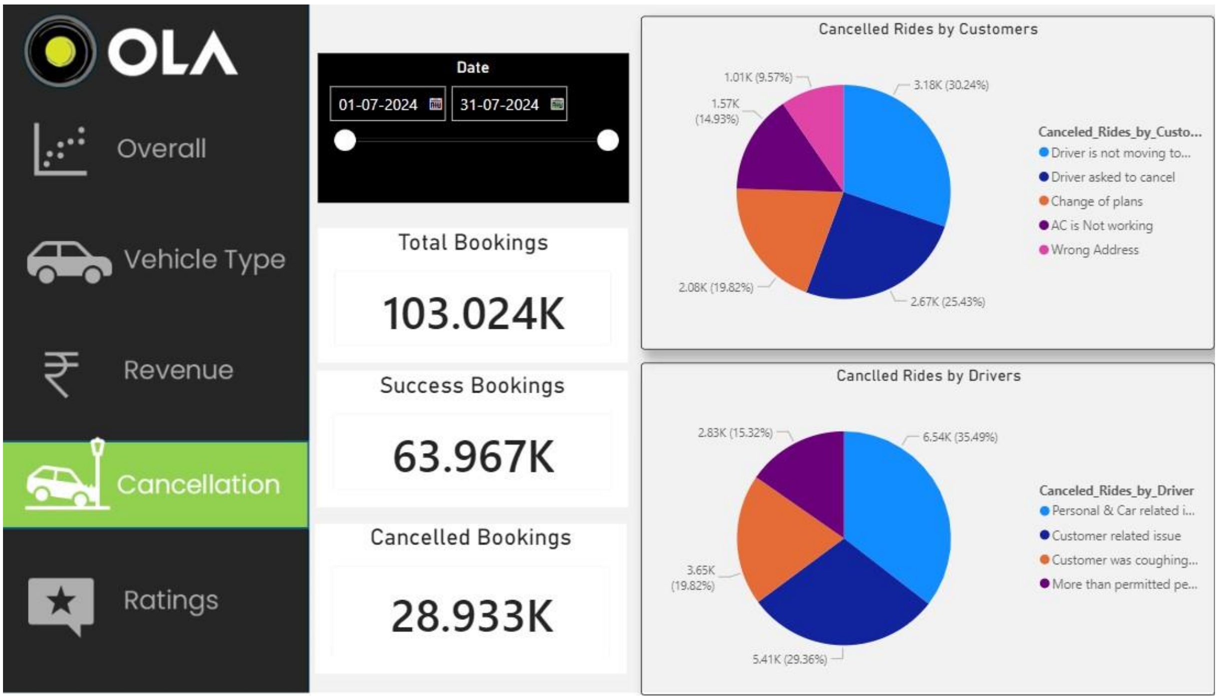
Shows booking value, successful bookings, and travel distance by vehicle type.

Revenue and Ride Distance Analysis



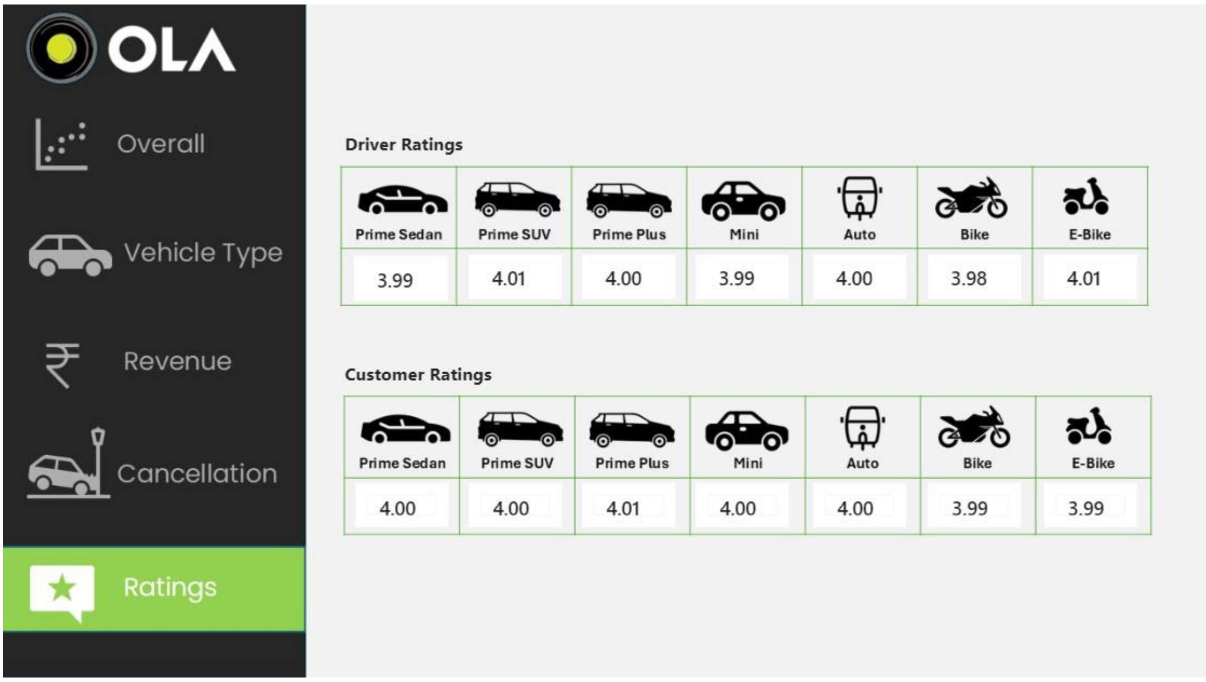
Displays revenue by payment methods and daily ride distance trends.

Cancellation Analysis Dashboard



Displays booking cancellations by customers, drivers, and overall booking status.

Driver and Customer Ratings Analysis



Driver and Customer Ratings Analysis

10. Project Outputs

- Cleaned Dataset (CSV)
- Jupyter Notebook (Analysis + Visualizations)
- Power BI Dashboard (.pbix)
- Insights Report

11. Links

GitHub Repository
https://github.com/abdulrahmann4111-spec/ola-project-/tree/main